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SHREE L. R. TIWARI COLLEGE OF ENGINEERING

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DEPARTMENT OF COMPUTER ENGINEERING

CML610 Cloud Computing Lab

Sixth Semester, 2025-2026 (Even Semester)

Name of Student : Mainuddin Khan

Roll No. : 53

Division :A

Batch :II

Day / Session : Monday /Friday

Venue : 310

Experiment No. :3

Title of Experiment : To study and Implement Bare-metal Virtualization using Xen, HyperV or VMware Esxi.

Date of Conduction :23/01/26

Date of Submission :

Particulars	Max. Marks	Marks Obtained
Preparedness and Efforts(PE)	**3**	
Knowledge of tools(KT)	**3**	
Debugging and results(DR)	**3**	
Documentation(DN)	**3**	
Punctuality & Lab Ethics(PL)	**3**	
Total	**15**	

Grades – Meet Expectations (3 Marks), Moderate Expectations (2 Marks), Below Expectations (1 Mark)

Checked and Verified by

Name of Faculty : Pravin Jangid

Signature :

Date :

Experiment 3

Aim:

To study and Implement Bare-metal Virtualization using Xen, HyperV or VMware Esxi.

Theory:

Hypervisor

A Hypervisor, also called a Virtual Machine Monitor (VMM), is a software layer that enables multiple virtual machines (VMs) to run simultaneously on a single physical system. It acts as an interface between the hardware and the virtual machines, controlling and allocating resources such as CPU, memory, storage, and network.

The hypervisor ensures that each virtual machine operates in isolation, meaning one VM does not interfere with another, even though they share the same physical hardware. This improves resource utilization, scalability, and cost efficiency.

Types of Hypervisors

1. Host Hypervisor :A Host Hypervisor is installed on top of an existing operating system such as Windows or Linux. The host operating system manages the hardware, while the hypervisor functions as an application that creates and runs virtual machines.

Architecture

Hardware → Host Operating System → Hypervisor → Virtual Machines

Working

- The host OS boots first.
- The hypervisor is installed as software.
- Virtual machines request hardware access through the host OS.

Advantages

- Easy to install and configure
- Suitable for beginners and experimentation
- Does not require dedicated hardware

Disadvantages

- Lower performance due to dependency on host OS
- Less secure than bare-metal hypervisors

Examples

- VirtualBox
- VMware Workstation

Usage

- Academic experiments
- Software testing
- Development environment

2. Bare Metal Hypervisor: A Bare Metal Hypervisor runs directly on the physical hardware without any host operating system. It directly manages hardware resources and provides virtual machines with better performance and security.

Architecture

Hardware → Hypervisor → Virtual Machines

Working

- Hypervisor is installed directly on the server.
- It controls CPU scheduling, memory allocation, and I/O operations.
- Virtual machines communicate directly with the hypervisor.

Advantages

- High performance and low latency
- Better security and stability
- Efficient resource utilization

Disadvantages

- Complex installation
- Requires technical expertise

Examples

- VMware ESXi
- XEN
- Microsoft Hyper-V

Usage

- Data centers
- Cloud computing
- Enterprise virtualization

XEN Hypervisor :

XEN is an open-source Type-1 (bare-metal) hypervisor that provides high performance and scalability. It is widely used in cloud platforms like Amazon Web Services (AWS).

Architecture of XEN

- **XEN Hypervisor** runs directly on hardware
- **Domain 0 (Dom0):** A privileged control domain that manages hardware and virtual machines
- **Domain U (DomU):** Unprivileged guest virtual machines

Features

- Supports **paravirtualization** and **full virtualization**
- Strong isolation between virtual machines
- High security and scalability

Advantages

- Open source and flexible
- Efficient performance
- Widely used in cloud environments

VMware ESXi :

VMware ESXi is a commercial Type-1 hypervisor developed by VMware. It is designed for enterprise-level virtualization and runs directly on server hardware.

Architecture of ESXi

- VMkernel: Core component that manages CPU, memory, and storage
- Direct access to hardware without host OS
- Managed using **vSphere Client**

Features

- High performance and reliability
- Advanced security features
- Supports large number of virtual machines

Advantages

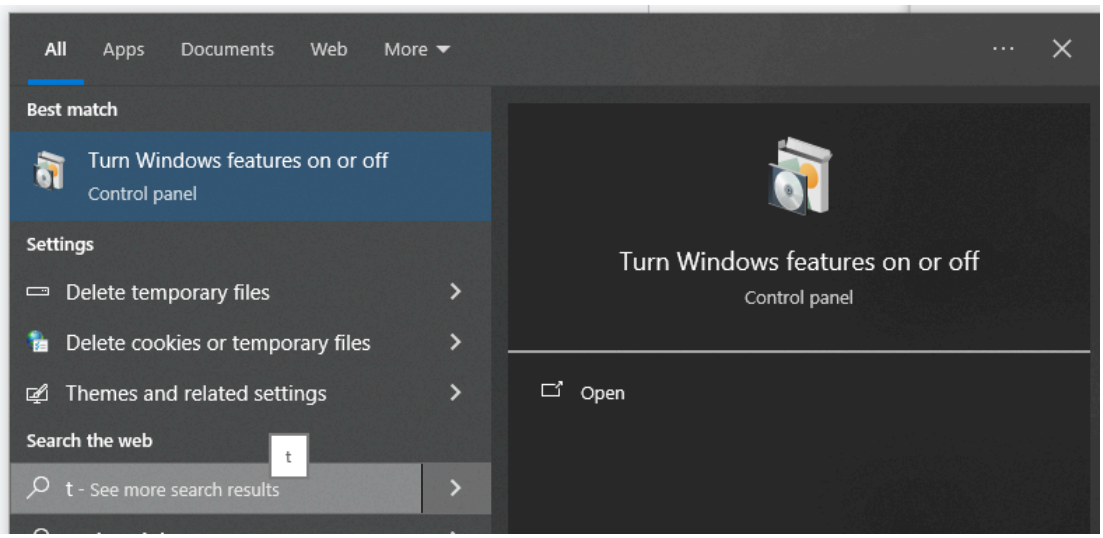
- Enterprise-grade stability
- Efficient resource management
- Widely used in data centers

Installation of Operating System on Virtual Machine

Steps to Install OS on VM using Hyper-V

Step 1: Enable Hyper-V

1. Open **Control Panel**
2. Click **Programs** → **Turn Windows features on or off**
3. Check **Hyper-V**
4. Click **OK** and restart the system

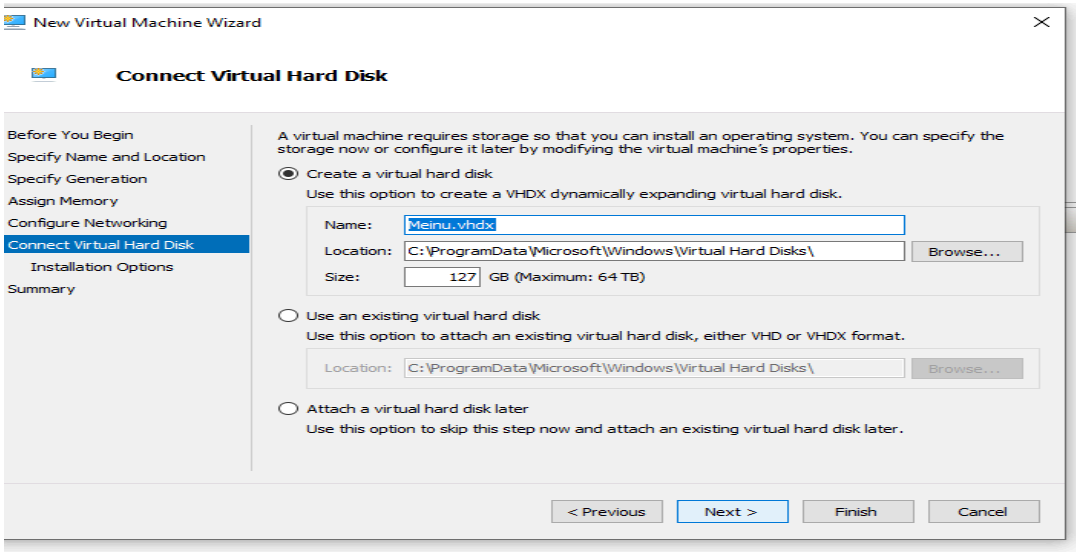


Step 2: Open Hyper-V Manager

1. Press **Windows + S**
2. Search **Hyper-V Manager**
3. Open it
4.

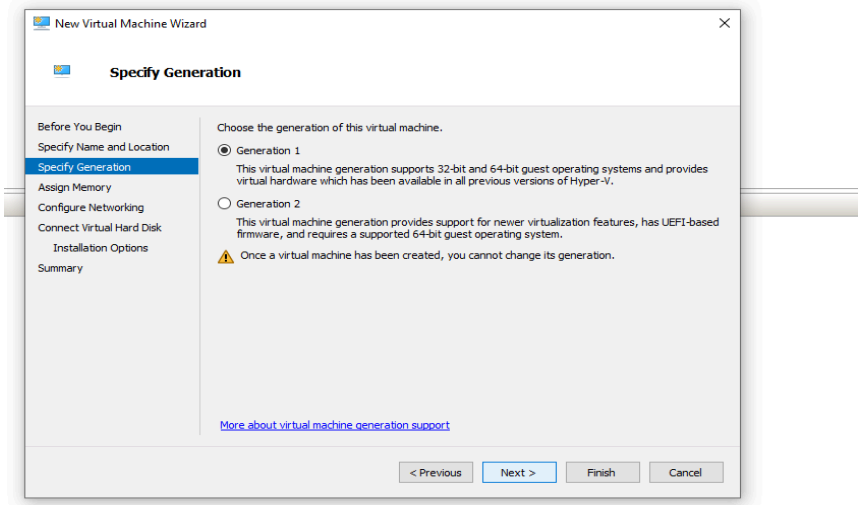
Step 3: Create a New Virtual Machine

1. In Hyper-V Manager, click **New** → **Virtual Machine**
2. Click **Next**
3. Enter **VM Name**
4. Choose location (optional)



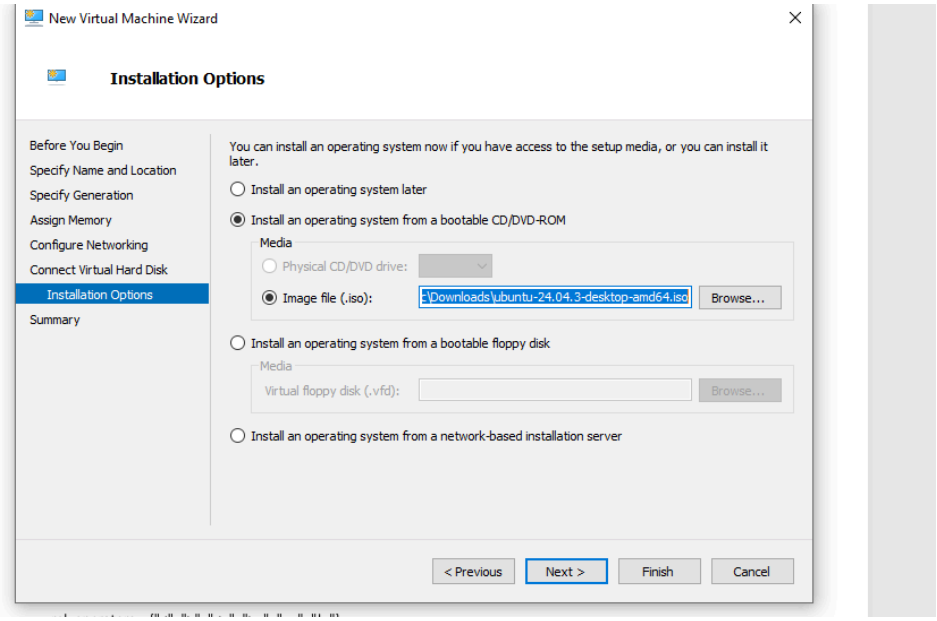
Step 4: Choose VM Generation

- Select **Generation 1** (for older OS)
- Select **Generation 2** (recommended for modern OS)



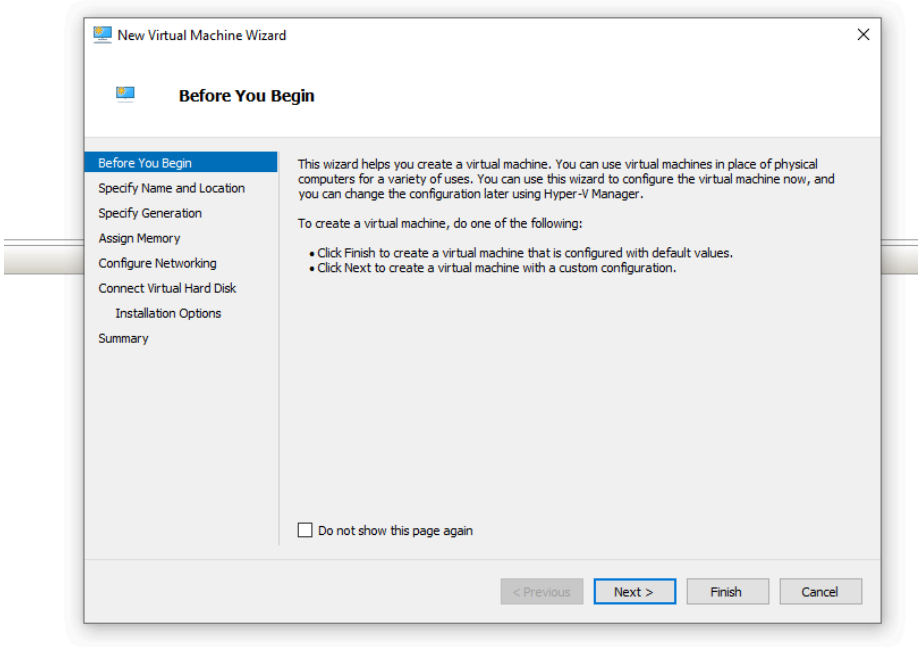
Step 5: Assign Memory

- Allocate **RAM** (e.g., 2048 MB or more)
- Enable **Dynamic Memory** (optional)



Step 6: Configure Network

- Select **Default Switch** or available virtual switch

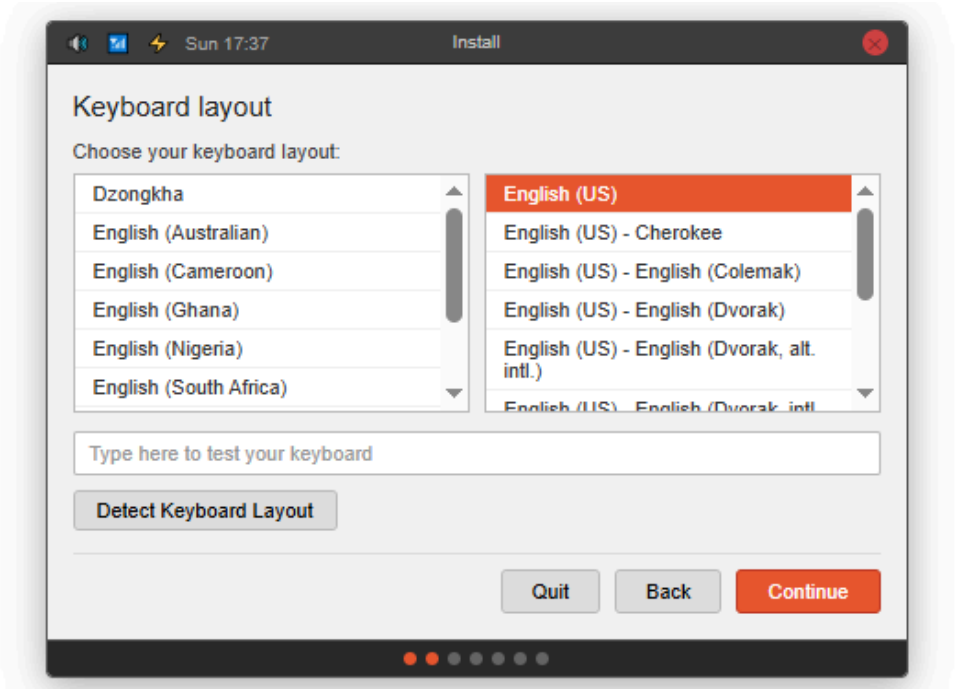


Step 7: Attach Virtual Hard Disk

- Create a new virtual hard disk
- Specify disk size (e.g., 40 GB)

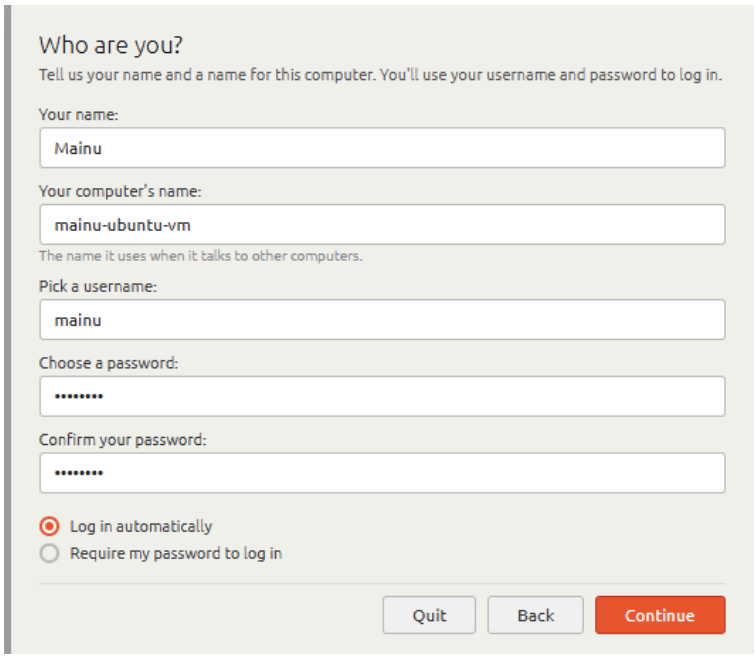
Step 8: Attach OS ISO File

1. Select **Install OS from image file**
2. Browse and select **ISO file**
3. Click **Finish**



Step 9: Start VM and Install OS

1. Right-click VM → **Connect**
2. Click **Start**
3. OS installation screen appears
4. Follow on-screen installation steps



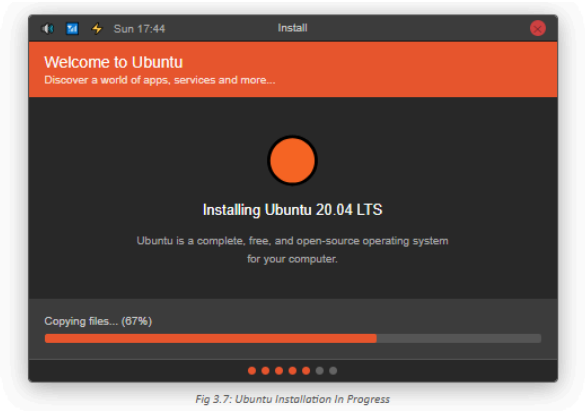
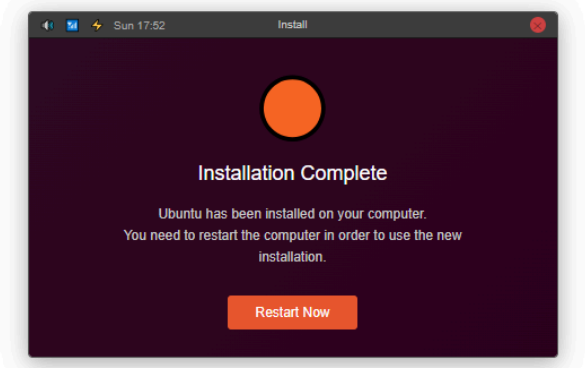


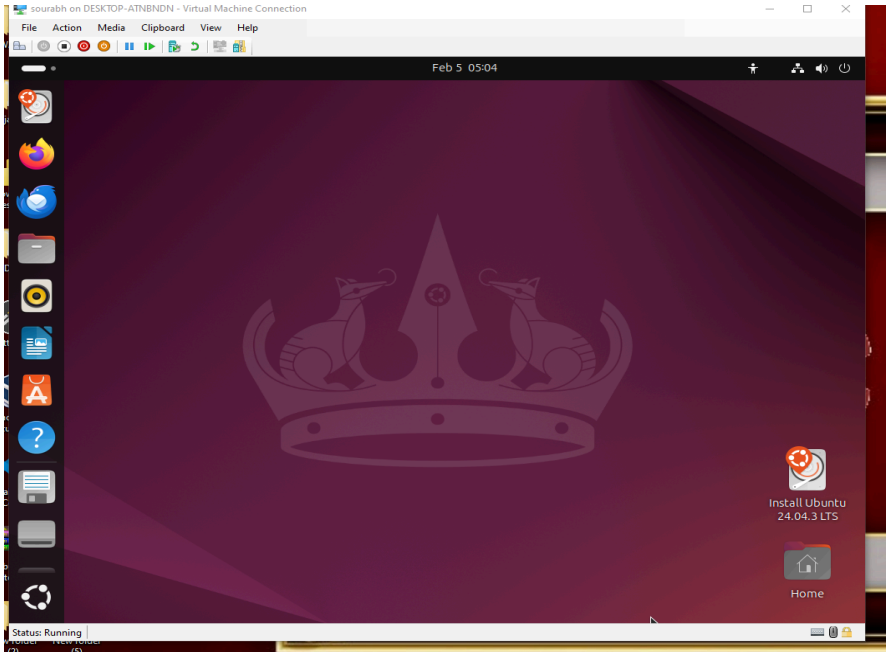
Fig 3.7: Ubuntu Installation In Progress

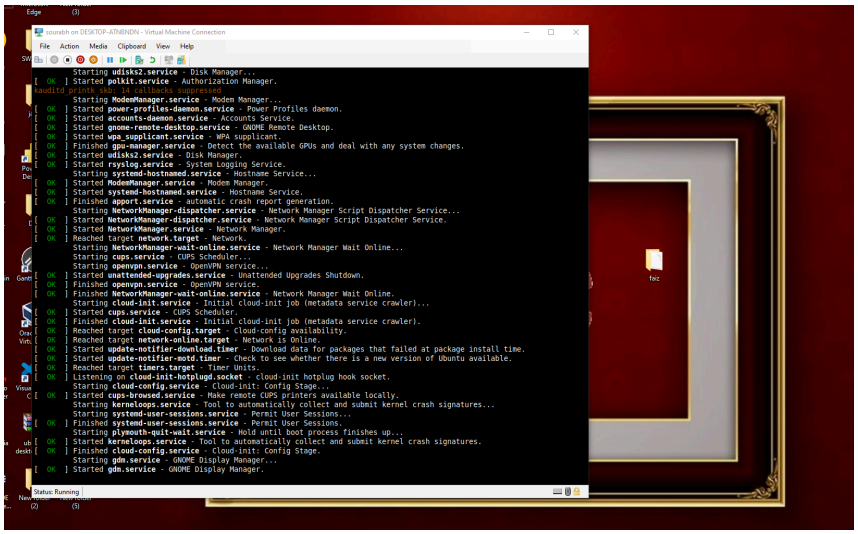
8. When installation is complete, click **Restart Now**. Remove the installation media when prompted, then press **Enter**.



Step 10: OS Installation Completed

- VM restarts
- OS login screen appears





CONCLUSION

Virtualization allows multiple operating systems to run on a single physical machine using hypervisors. Host hypervisors are suitable for learning and testing, while bare-metal hypervisors

provide better performance, security, and reliability for enterprise use. Virtualization improves resource utilization and reduces hardware cost.

Criteria to Select a Hypervisor

- **Purpose** – Testing or production use
- **Performance** – High performance needs Type-1 hypervisor
- **Security** – Type-1 offers better isolation
- **Hardware support** – Server or personal system
- **Cost** – Open-source or commercial solution

How to Resolve Limitations of Virtualization

- Enable hardware virtualization (VT-x / AMD-V)
- Use bare-metal hypervisors for better performance
- Allocate resources properly to VMs
- Apply security updates regularly
- Use fast storage and efficient network setup